AI TRAINING – WEEK 2 ASSESSMENT

NumPy + Pandas Checkpoint

Timed Data Manipulation Tasks + 20 MCQs + EDA Submission

1. Assessment Overview

This checkpoint checks your understanding of NumPy and Pandas before moving to more advanced data analysis.

- Part A – Timed MCQ quiz: 20 questions, 20 minutes, 20 marks
- Part B – Timed data manipulation: 4 tasks, 40 minutes, 40 marks
- Part C – Debugging and explanation: 10 marks
- Part D – Week 2 notebooks and EDA submission: 30 marks

Suggested total: 100 marks. Trainer may adjust the timing.

2. Student Instructions

- Complete the timed quiz independently. Do not use Google, ChatGPT, or other AI tools during the timed section.
- Write and run your own code for practical tasks.
- Include outputs or a short note confirming results.
- If your result is incorrect, record the error and explain the fix.
- Use clear variable names and organize your notebook.
- Do not copy the trainer answer key into your submission.

3. Part A – Timed MCQ Quiz (20 Questions)

Google Forms suggestion: 20 minutes, 1 mark per question. Trainer answer key is at the end.

1. Which statement correctly creates a NumPy array?

- A. arr = numpy(1, 2, 3)
- B. arr = np.array([1, 2, 3])
- C. arr = array(1, 2, 3)
- D. arr = np.list([1, 2, 3])

2. What does arr.ndim return?

- A. Number of elements
- B. Number of dimensions
- C. Data type
- D. Number of rows only

3. If arr = np.array([10, 20, 30]), what is arr[-1]?

- A. 10
- B. 20
- C. 30
- D. Error

4. What does arr.shape describe?

- A. Sizes along each dimension
- B. Only the data type
- C. Sum of values
- D. Maximum value

5. For same-shaped arrays a and b, what does a * b do?

- A. Matrix multiplication
- B. Element-wise multiplication
- C. Adds arrays
- D. Joins arrays

6. Which operator commonly performs matrix multiplication?

- A. *
- B. +
- C. @
- D. %

7. What is vectorization?

- A. Convert values to strings
- B. Apply operations to arrays without an explicit Python loop
- C. Sort a list
- D. Remove duplicate rows

8. What does np.mean(arr) calculate?

- A. Median
- B. Average
- C. Maximum
- D. Number of dimensions

9. Which import is conventional for Pandas?

- A. import pandas as pd
- B. import pandas as np
- C. from pandas import csv
- D. import pd

10. What does pd.read_csv("sales.csv") usually return?

- A. List
- B. DataFrame
- C. String
- D. NumPy scalar

11. What does df.head() show by default?

- A. Last five rows
- B. First five rows
- C. Column names only

- D. Summary statistics only

12. Which method shows columns, non-null counts, and data types?

- A. df.describe()
- B. df.info()
- C. df.tail()
- D. df.shape()

13. What does df.shape return?

- A. Tuple of row and column counts
- B. Missing-value DataFrame
- C. Column-name list
- D. Column averages

14. What is df["Sales"] generally?

- A. DataFrame with all columns
- B. Series
- C. Python package
- D. CSV file

15. Which filters rows where Sales > 100?

- A. df("Sales" > 100)
- B. df[df["Sales"] > 100]
- C. df["Sales" > 100]
- D. df.filter(Sales > 100)

16. Which symbol combines Pandas conditions as element-wise AND?

- A. and
- B. &&
- C. &
- D. AND

17. When combining Pandas conditions with &, what should you do?

- A. Use curly braces
- B. Put each condition in parentheses
- C. Convert DataFrame to string
- D. Use only one condition

18. Which checks missing values by column?

- A. df.isna().sum()
- B. df.head().sum()
- C. df.shape.isna()
- D. df.columns.missing()

19. Which counts fully duplicated rows?

- A. df.duplicates()
- B. df.duplicated().sum()

- C. df.drop().sum()
- D. df.repeat().count()

20. Why inspect a dataset before analysis?

- A. Make file larger
- B. Understand structure, types, and data quality
- C. Remove every text column
- D. Avoid writing code

4. Part B – Timed Data Manipulation Tasks

Suggested duration: 40 minutes. Use a Jupyter Notebook and show code/results.

Task 1 – NumPy Array Basics (10 marks)

Create an array containing [12, 18, 25, 31, 44, 50].

- Print shape, size, and number of dimensions.

```
import numpy as np
array=np.array([12, 18, 25, 31, 44, 50])
print(array.size)
print(array.shape)
print(array.ndim)
```

output;

6
(6,)
1

- Print frst and last elements.

```
import numpy as np
array=np.array([12, 18, 25, 31, 44, 50])
print(array[0],array[-1])
```

ouptu:
12 50

- Use slicing to select 18, 25, and 31.

```
import numpy as np
array=np.array([12, 18, 25, 31, 44, 50])
print(array[1:4])
```

output: [18 25 31]

- Add 10 to every element using vectorization.

```
import numpy as np
array=np.array([12, 18, 25, 31, 44, 50])
new=array+10
print(new)
```

output: [22 28 35 41 54 60]

- Calculate mean, maximum, and minimum.

```
import numpy as np
array=np.array([12, 18, 25, 31, 44, 50])
print(np.mean(array))
print(min(array))
print(max(array))
```

output:
30.0
12
50

Task 2 – NumPy Filtering and Matrix Operations (10 marks)

Create scores = np.array([45, 67, 82, 39, 91, 74]).

- Select scores greater than or equal to 70.

```
import numpy as np
scores = np.array([45, 67, 82, 39, 91, 74])
score=scores[scores>=70]
print(score)
```

output: [82 91 74]

- Count passing scores if 40 is the pass mark.

```
import numpy as np
scores = np.array([45, 67, 82, 39, 91, 74])
count=0
for score in scores:
    if score>40:
        count+=1
print(f'total {count} are passed')
```

output: total 5 are passed

- Create two 2×2 arrays and show element-wise multiplication and matrix multiplication.

```
import numpy as np
a = np.array([[1, 2],
              [3, 4]])
b = np.array([[5, 6],
              [7, 8]])
```

```
# ########Element-wise multiplication###########
print("Element-wise multiplication:")
print(a * b)
######### matrix multiplication  #########
print("Matrix multiplication:")
print(a @ b)
```

- Explain how * differs from @.
  Ans:when we are multiplcated two 2*2 arrays using element wise multiplication then the elements will muliplicate which is at same position. But matrix multiplication is not like that Multiplies rows by columns and adds.

Task 3 – Pandas Loading, Inspection, and Filtering (10 marks)

Create a DataFrame:
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North", "South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120, 30, 60]}
df = pd.DataFrame(data)

- Display head(), info(), and descriptive statistics.

```
import pandas as pd
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North", "South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120, 30, 60]}
df = pd.DataFrame(data)
print(df.describe)
print(df.info())
print(df.head())
```

```
              Sales      Profit
count     5.000000    5.000000
mean    320.000000   56.000000
std     279.732015   39.115214
min     100.000000   20.000000
25%     150.000000   30.000000
50%     250.000000   50.000000
75%     300.000000   60.000000
max     800.000000  120.000000

<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4

Data columns (total 4 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Product  5 non-null      str
 1   Region   5 non-null      str
 2   Sales    5 non-null      int64
 3   Profit   5 non-null      int64
dtypes: int64(2), str(2)
memory usage: 292.0 bytes
```

None

	Product	Region	Sales	Profit
0	Pen	South	100	20
1	Book	North	250	50
2	Bag	South	800	120
3	Pen	West	150	30
4	Book	West	300	60

- Select Sales and confirm its type.

```
import pandas as pd
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North",
"South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120,
30, 60]}
df = pd.DataFrame(data)
print(df["Sales"].dtype)
```

output: int64

- Filter records where Sales > 200.

```
import pandas as pd
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North",
"South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120,
30, 60]}
df = pd.DataFrame(data)
print(df[df["Sales"]>200])
```

output:

	Product	Region	Sales	Profit
1	Book	North	250	50
2	Bag	South	800	120
4	Book	West	300	60

- Filter records where Sales > 200 and Profit > 50.

```
import pandas as pd
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North",
"South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120,
30, 60]}
df = pd.DataFrame(data)
print(df[(df["Sales"]>200) &(df["Profit"]>50)])
```

output:

	Product	Region	Sales	Profit
2	Bag	South	800	120
4	Book	West	300	60

- Calculate total sales and average profit.

```
import pandas as pd
data = {"Product": ["Pen", "Book", "Bag", "Pen", "Book"], "Region": ["South", "North",
"South", "West", "West"], "Sales": [100, 250, 800, 150, 300], "Profit": [20, 50, 120,
30, 60]}
```

```
df = pd.DataFrame(data)
print(df["Sales"].sum())
print(df["Profit"].mean())
```

output: 1600
56.0

Task 4 – EDA Business Questions (10 marks)

Using the DataFrame above, answer:

- Which product has the highest total sales?
  Ans:bag
- Which region has the highest total sales?
  Ans:south
- How many records have Sales > 200?
  Ans:3
- What is the average Sales value?
  Ans:320.0
- Are there missing values or duplicate rows?
  Ans:no missing values or duplicate rows.

5. Part C – Debugging and Explain Your Fix (10 marks)

Fix each issue and write one sentence explaining the cause.

Debugging Example 1

```
import numpy as np
arr = np.array([5, 10, 15])
print(arr[3])
```

output:array index out of bounce because it is having only three elements but it is accesing for 4th element.

Sol:

```
import numpy as np
arr = np.array([5, 10, 15])
print(arr[2])
```

Debugging Example 2

```
import pandas as pd
df = pd.DataFrame({"Sales": [100, 200, 300]})
print(df[(df["Sales"] > 100) & (df["Sales"] < 300)])
```

ans:there is no bug it will print 200 because it is in between remaining two values.

Debugging Example 3

```
import pandas as pd
df = pd.DataFrame({"Sales": ["100", "200", "300"]})
print(df["Sales"].mean())
```

ans:we cannot find mean for strings.mean is only used for numerical values.

Sol:

```
import pandas as pd
df = pd.DataFrame({"Sales": [100, 200, 300]})
print(df["Sales"].mean())
```

6. Part D – Week 2 Assessment + EDA Submission

- Week 2 NumPy practice: arrays, dimensions, indexing, slicing, vectorization, and matrix operations.
- Pandas checkpoint: Series/DataFrame, read_csv, head/info/describe, and filtering.
- EDA Notebook v1: a real CSV dataset, inspection, filtering, data-quality checks, and answers to 15 business questions.
- Include dataset name/source, code, outputs, and short explanations.
- Ensure each notebook runs from top to bottom.

7. Mistake Review

Record at least three mistakes or confusing concepts after the checkpoint.

Mistake / Question	What I Did	Correct Approach	What I Learned

8. Explain in Your Own Words

1. What is the difference between a NumPy array and a Pandas DataFrame?

Answer: numpy array is used for numerical conditions but pandas DataFrame is used analyze tabular form data.

2. What is vectorization, and why is it useful?

Answer: vectorization means writing code without using loops.it is useful to make code fatser to execute.

3. What is the difference between a Series and a DataFrame?

Answer: series is one datastructural form and data frame is two datastructural form

4. What do head(), info(), and describe() tell you?

Answer: head() gives us top 5 rows of data in dataset . info() give us coloumn names and data types non_null counts.describe() gives statiscal data.

5. How do you filter rows using two conditions?

Answer: use & for AND . “|” for OR operations for two conditions.

6. What data-quality issue did you find in your EDA dataset?

Answer: I checked for missing values and duplicate rows using `df.isna().sum()` and `df.duplicated().sum()`.

9. Student Submission Checklist

- I completed all 20 MCQs.
- I attempted all four practical tasks.
- I fixed the debugging examples and explained the fixes.
- I reviewed at least three mistakes.
- I submitted the NumPy practice notebook.
- I submitted EDA Notebook v1 with 15 business questions answered.
- I can explain the concepts without memorizing definitions.

10. Trainer Answer Key – Keep Separate from Student Quiz

MCQ answers: 1-B | 2-B | 3-C | 4-A | 5-B | 6-C | 7-B | 8-B | 9-A | 10-B | 11-B | 12-B | 13-A | 14-B | 15-B | 16-C | 17-B | 18-A | 19-B | 20-B

Suggested grading: MCQs 20 marks; practical tasks 40 marks; debugging/explanation 10 marks; Week 2 notebooks and EDA submission 30 marks. Total: 100 marks.

Key Principle: A good data analyst checks the data, validates the result, and explains what they found.